

Exact Smith Invariants and Affine Determinant Lines of Binary-Form Factorisation

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Abstract

For positive degrees $\mathbf{d}$, put $g = \gcd_i d_i$ and $\mathbf{e} = \mathbf{d}/g$, so that $\mathbf{e}$ is primitive. Pairwise resultants of labelled binary-form factors have characters $\chi_{ij} = [e_j \varepsilon_i + e_i \varepsilon_j]$ in $M = \mathbb{Z}^k / \mathbb{Z}1$. Let $W_E(\mathbf{e})$ be the matrix with these rows and let $h(\mathbf{e})$ be the positive gcd of its maximal minors. We compute the full quotient by these characters. For every prime p , choose a with $p \nmid e_a$. Then

$$\frac{M}{\langle \chi_{ij} \rangle} \otimes \mathbb{Z}_{(p)} \cong \frac{\mathbb{Z}_{(p)}}{(e_a - \sum_{i \neq a} e_i, 2e_i e_j : i < j, i, j \neq a)}.$$

This one-generator presentation gives every valuation of the final Smith invariant, including the full 2-adic case, a symmetric global formula, and an $O(k^2)$ gcd computation. It also shows that spanning-tree minors already generate the full maximal-minor gcd. Consequently the complete character matrix for the original degrees has Smith form $\text{diag}(g, \dots, g, gh(\mathbf{e}))$, with $k - 2$ copies of g .

We derive this lattice from an affine determinant-line theorem. Every maximal minor of the differential of $(A_1, \dots, A_k) \mapsto \prod_i A_i$ equals, over $\mathbb{Z}$, the product of all pairwise resultants times the corresponding Plücker coordinate of the product-one scaling torus, with an explicit orientation. Multi-bordering contracts this tensor with a vertical torus Jacobian; for semi-invariants that Jacobian is an integer character determinant. On the universal pairwise-coprime open, pairwise resultants generate all regular units modulo constants, so the complete lattice defines the intrinsic unit-character residual group of that open, not a classification of arbitrary semi-invariants. A square Laurent-unit normalisation realises its index. More generally, for a square full-rank semi-invariant character matrix W over an algebraically closed field, the normalised factorisation map is finite locally free on a dense target open of degree $|\det W|D!/\prod_i d_i!$; its separable and inseparable factors follow from the Smith entries of W .

The square multiplication Jacobian, multifactor scalar resultant, signed incidence support, arithmetic-matroid multiplicity, and labelled root-partition count are classical inputs. The candidate increment is the exact local/global calculation for the cross-weighted diagonal-quotient character list, together with its factorisation interpretation and the explicit integral affine Plücker lift. The literature search is bounded, and the exact SymPy and FLINT checks are producer-side regression evidence rather than independent reproduction.

Keywords: binary forms; resultants; Smith normal form; arithmetic matroids; torus characters; determinant lines; factorisation maps

Parent work. This paper is a separately versioned extension of the unrefereed candidate Bordered Jacobian Foundations; see the Evidence Press parent page and the immutable parent archive doi:10.5281/zenodo.21855302.

中文摘要

對正次數向量 $\mathbf{d}$ ，令 $g = \gcd_i d_i$ 及 $\mathbf{e} = \mathbf{d}/g$ ，故 $\mathbf{e}$ 為本原向量。標記二元齊次式各成對結式在特徵格 $\mathbb{Z}^k/\mathbb{Z}1$ 中給出交叉加權特徵 $\chi_{ij} = [e_j \varepsilon_i + e_i \varepsilon_j]$ 。令 $W_E(\mathbf{e})$ 為以這些特徵為列之矩陣，並以 $h(\mathbf{e})$ 表示其最大子式的正最大公因數。本文逐質數計算由全部 χ_{ij} 生成之格的商：選取 $p \nmid e_a$ 後，局部商為一生成元模，其關係理想由 $e_a - \sum_{i \neq a} e_i$ 與所有 $2e_i e_j$ ($i, j \neq a$) 生成。此表示給出最後 Smith 不變因子的每個 p -進賦值（包括完整的二進情形）、對稱的整體公式，以及僅用生成樹子式即可取得的最大子式公因數。因此未約化次數向量的 Smith 標準形為 $\text{diag}(g, \dots, g, gh(\mathbf{e}))$ 。

此格源自一個整數型仿射行列式線定理：多因子乘法微分的每一最大子式，皆為全部成對結式之乘積上乘積為一縮放環面的相應 Plücker 座標，並有明確定向。加入邊界函數後得到環面方向 Jacobian；對半不變量而言即為整數特徵行列式。在通用的成對互素開集上，成對結式在常數之外生成所有正則單位，故完整特徵格內在地描述該開集的「單位特徵」殘餘群，而非任意半不變量的分類。對代數閉域上的滿秩方形特徵矩陣，正規化因式分解映射在稠密目標開集上為有限局部自由，其次數為 $|\det W|D!/\prod_i d_i!$ ，可分與不可分部分由 Smith 不變因子決定。

方形乘法 Jacobian、多因子純量結式、有號關聯矩陣支撐、算術擬陣乘數與標記根分割皆屬經典輸入。本文候選增量為上述交叉加權對角商特徵格的精確局部與整體計算、其因式分解詮釋，以及明確的整數型仿射 Plücker 提升。文獻檢索範圍有限；SymPy 與 FLINT 的精確檢查僅屬作者端回歸證據，並非獨立重現。

關鍵詞：二元齊次式；結式；行列式線；代數環面；Smith 標準形；因式分解映射

1. Introduction

Let V_d be the affine coefficient space of binary forms of degree d . For two positive degrees r, s , coefficient multiplication is the morphism

$$m_{r,s} : V_r \times V_s \longrightarrow V_{r+s}, \quad (A, B) \longmapsto AB.$$

Its source dimension exceeds its target dimension by one. The generic kernel of its differential is the infinitesimal scaling direction $(A, -B)$, induced by $(A, B) \mapsto (\lambda A, \lambda^{-1} B)$. The parent result proves an exact integral refinement of this dimension count: the complete signed maximal-minor vector of $Dm_{r,s}$ is a fixed sign times $\text{Res}(A, B)(A, -B)$ (Anonymous 2026). A border covector pairs with this missing direction. When the border is the differential of a bihomogeneous function, the pairing is its scaling-degree difference. Appendix E includes a self-contained proof of that base identity in the present coefficient and Sylvester conventions.

The rank-one vector identity is a determinant-line completion of the classical square chart identities reviewed below. It is used here as the base of an explicit all-factor construction, not as a standalone priority claim.

The rank-one formula suggests a more general object. Given positive degrees $d_1, \dots, d_k$, multiplication

$$m_d : \prod_{i=1}^k V_{d_i} \longrightarrow V_D, \quad D = \sum_i d_i,$$

has relative dimension $k - 1$. The product-one torus

$$T = \{(\lambda_1, \dots, \lambda_k) \in \mathbb{G}_m^k : \lambda_1 \cdots \lambda_k = 1\}$$

acts within every fibre. Pairwise common roots are detected by the resultants $R_{ij} = \text{Res}(A_i, A_j)$. The natural candidate for the determinant line is therefore a product of the pairwise-collision divisor and the Plücker line of the torus tangent space.

Several exact shadows of this expectation are classical. For two factors, coefficient-product equations have a square Jacobian equal, up to the stated coordinate convention, to a Sylvester matrix and hence to the resultant; this appears in Hensel-lifting, real-algebraic-geometry, projective-tangent, and arithmetic settings (Artin 2022, sec. 1.8, Lemma 1.8.5; Basu et al. 2006, sec. 4.2.1; Chipalkatti 2003, Lemma 5.7; Bhargava et al. 2022, Lemma 2.6). For an arbitrary number of monic factors, Chaperon and López de Medrano give a corank formula and prove that pairwise coprimality is equivalent to local invertibility (Chaperon and López de Medrano 2009, Theorem 3). Mahatab and Sampath prove that the determinant of the canonical polynomial Chinese remainder map is the product of the pairwise resultants (Mahatab and Sampath 2015, Theorem A.3). Generalized Sylvester and Koszul-complex constructions also place the resultant in the relevant determinantal ideals; Chardin proves divisibility of the maximal minors and a generic gcd statement (Chardin 1993, sec. IV, Lemma 1 and Remark 1). Projective polynomial multiplication and its finite root-partition fibres are classical (Breiding et al. 2024), as are factorisation quotients by scaling tori in the linear-factor case (Kurth 1997). These antecedents determine the claim boundary. We do not claim the square product-Jacobian identity, the monic multi-factor rank criterion, the scalar product $\prod R_{ij}$, projective root partitions, or incidence minors as new.

The principal arithmetic object is the complete edge-character list. Put $g = \gcd_i d_i$ and $\mathbf{e} = \mathbf{d}/g$. In the character lattice

$$M = X^*(T) = \mathbb{Z}^k / \mathbb{Z}\mathbf{1},$$

the primitive degrees $\mathbf{e}$ give the rows of a matrix $W_E(\mathbf{e})$:

$$\chi_{ij} = [e_j \varepsilon_i + e_i \varepsilon_j], \quad C_{\mathbf{e}} = M / \langle \chi_{ij} : i < j \rangle.$$

Let $h(\mathbf{e})$ be the positive gcd of the maximal minors of $W_E(\mathbf{e})$. Section 7 proves that it equals $|C_{\mathbf{e}}|$.

The main theorem computes $C_{\mathbf{e}}$ locally at every prime by a one-generator presentation. It follows that $C_{\mathbf{e}}$ is cyclic, that every $v_p(|C_{\mathbf{e}}|)$ has a closed formula, and that the final index has a symmetric factorisation-free expression. For $\mathbf{d} = g\mathbf{e}$, this yields the complete Smith form

$$\text{diag}(\underbrace{g, \dots, g}_{k-2}, gh(\mathbf{e})).$$

The proof chain leading to this arithmetic calculation has four further parts. First, an integral affine lift packages every maximal minor of $Dm_{\mathbf{d}}$ as $\Delta_{\mathbf{d}} = \prod_{i < j} R_{ij}$ times the matching torus Plücker coordinate, with a fixed orientation. Second, Laplace expansion and Cauchy – Binet give the multi-border formula; semi-invariant borders contribute $\det(W) \prod_a g_a$. Third, resultant characters reduce integrally to a cross-weighted all-negative incidence problem, giving all square character minors and showing that spanning trees alone determine the full-list index. Fourth, pairwise resultants are proved to generate every regular unit modulo constants on the universal pairwise-coprime open. Hence the complete lattice is intrinsic in that regular-unit category, and its quotient is the unit-character residual group of that open.

The contribution hierarchy is intentionally explicit.

Layer	Status used in this paper
Square multiplication Jacobian, multifactor scalar resultant, monic rank criterion, and labelled root partitions	Classical antecedents
All-factor affine Plücker tensor and multi-border identity	Explicit integral lift and formal contraction of classical scalar/kernel data; candidate coordinate-level contribution
Signed support and powers of two in character minors	Classical all-negative signed-incidence theory
Degree weights, appended balance row, and resultant interpretation	Factorisation-specific specialisation
$m(E)$, the maximal-minor gcd, and complex toric component count	Standard arithmetic-matroid and toric-arrangement data
Local cyclic presentation, exact valuations, global formula, and scaled Smith form for the cross-weighted diagonal-quotient list	Strongest candidate theorem package in this paper
Generic degree $ \det W D!/\prod_i d_i!$	Scheme-theoretic synthesis of a labelled root-partition cover and a torus isogeny

For a square full-rank semi-invariant character matrix W over an algebraically closed field, we prove after restriction to an explicit dense target open that the normalised factorisation map is finite locally free of degree

$$|\det W| \frac{D!}{\prod_i d_i!}.$$

Smith form then separates its total, separable, and inseparable degrees. A square Laurent basis of the full regular-unit character lattice always realises the full-list index, but a square chart made from individual resultants need not do so. The results do not classify arbitrary polynomial semi-invariants, recognise level sets as affine spaces, classify Keller maps, or imply a Hessian conjecture result. The exact programs are producer-side regression checks; they are not independent reproduction, formal verification, or peer review.

2. Conventions and antecedents

2.1 Coefficients and resultants

For a positive integer d , write

$$A = \sum_{j=0}^d a_j X^{d-j} Y^j.$$

After dehomogenising by $X = 1$, the coefficient vector $(a_0, \dots, a_d)$ is read in ascending powers of Y . For forms of degrees r and s , the Sylvester matrix consists of the s shifted rows of the first coefficient vector followed by the r shifted rows of the second. In this convention, $\text{Res}(X^r, Y^s) = 1$. This choice fixes every sign below.

The classical interpretation of a resultant as a determinant goes back to Cayley (Cayley 1848) and is developed through modern determinant-of-complexes formalisms by Jouanolou, Chardin, Demazure, and Gelfand – Kapranov – Zelevinsky (Jouanolou 1991; Chardin 1993; Demazure 2012; Gelfand et

al. 1994). Jouanolou’s work also contains a bordered-minor evaluation for generic linear forms and extensive gradient and bordered-Macaulay identities (Jouanolou 1991, secs. 5.4.1 – 5.4.4; 1995; 1997). Those identities concern square elimination systems, resultant complexes, or congruences modulo the resultant. The morphism studied here differentiates multiplication with respect to factor coefficients and is rectangular before borders are chosen.

Adjacent work on binary discriminants and resultants includes the perfectness and equivariant resolutions of their Jacobian ideals (D’Andrea and Chipalkatti 2007). That theory concerns the resultant hypersurface and its coefficient derivatives. More directly, Chipalkatti proves that the differential of projective two-factor binary-form multiplication is injective exactly for coprime factors (Chipalkatti 2003, Lemma 5.7), while Chaperon and López de Medrano analyse the corank and singularities of multiplication of monic polynomials, including arbitrary numbers of factors (Chaperon and López de Medrano 2009, Theorems 1 – 4). These sources establish the rank-locus background. They do not, in the expanded bounded corpus examined for this project, state the universal non-monic all-factor complementary-minor tensor with its integral orientation.

2.2 Dimensions and the ordered torus basis

Fix $k \geq 2$ and positive degrees $\mathbf{d} = (d_1, \dots, d_k)$. Set

$$D = \sum_{i=1}^k d_i, \quad n = D + 1, \quad t = k - 1, \quad \text{quad} N = D + k = n + t.$$

The differential $M = Dm_{\mathbf{d}}$ is an $n \times N$ matrix. Source coordinates are ordered first by factor and then by coefficient. For $b = 1, \dots, t$, define the source tangent vector

$$\kappa_b = (0, \dots, 0, A_b, 0, \dots, 0, -A_k),$$

where the two nonzero blocks are the b -th and last factor blocks. Let K be the $t \times N$ matrix with rows κ_b . Direct differentiation gives

$$MK^T = 0.$$

These rows correspond to the cocharacter basis $e_b - e_k$ of the product-one torus.

For a t -element subset $I \subset \{0, \dots, N - 1\}$, let $M_{\hat{I}}$ be the square matrix obtained by deleting the columns in I . Define

$$p_I(M) = (-1)^{\sum_{i \in I} i} \det M_{\hat{I}}.$$

The zero-based sign is part of the theorem, not shorthand for an unspecified orientation. Write K_I for the $t \times t$ submatrix of K in the columns I .

Finally, define the pairwise-collision product

$$\Delta_{\mathbf{d}} = \prod_{1 \leq i < j \leq k} \text{Res}(A_i, A_j).$$

2.3 Square and monic shadows

Fixing one leading coefficient and the product leading coefficient removes a two-factor affine scaling direction and turns the remaining product equations into a square Sylvester map. Artin identifies this Jacobian directly (Artin 2022, sec. 1.8, Lemma 1.8.5); Basu, Pollack, and Roy give the monic version

(Basu et al. 2006, sec. 4.2.1), and Bhargava, Cremona, Fisher, and Gajović prove over an arbitrary ring that the monic-times-arbitrary chart has Jacobian equal to the resultant (Bhargava et al. 2022, Lemma 2.6). The Stacks Project uses the same Sylvester Jacobian to describe the coprime monic chart as étale (The Stacks Project Authors 2026, tag 00U0). For several monic factors, the product of pairwise resultants is also supplied by the polynomial Chinese-remainder determinant (Mahatab and Sampath 2015, Theorem A.3).

Thus neither the square Jacobian nor the multifactor scalar is a contribution of this paper. Before any leading coefficient is fixed, however, the differential has a $(k - 1)$ -dimensional kernel. The refinement recorded below retains the full complementary-minor tensor: which maximal minor corresponds to which torus Plücker coordinate, with what universal integral orientation. This lift is close to the classical identities and is formal from a generic determinant-line viewpoint once one chart minor and the kernel are known. The value of the treatment below is its universal coordinate identity, integral orientation, and compatibility with all complementary minors. No standalone priority claim is made for the lift. The next section isolates the composition rule used here.

3. A determinant-line composition lemma

The induction factors k -fold multiplication through $(k - 1)$ -fold multiplication and one two-factor multiplication. The following complementary-minor identity works over an arbitrary commutative ring, so no division by a selected nonzero minor or generic-rank reduction is needed.

Lemma 3.1 (block-composition sign). Let R be a commutative ring and fix $r, q \geq 1$ and $t_1 \geq 0$. Put

$$m = r + q - 1, \quad N = r + t_1 + q,$$

and suppose

$$P \in R^{r \times (r+t_1)}, \quad L \in R^{m \times (m+1)}, \quad S = \begin{pmatrix} P & 0 \\ 0 & I_q \end{pmatrix} \in R^{(m+1) \times N}.$$

Set $M = LS$. Assume that $K_P \in R^{t_1 \times (r+t_1)}$, $\beta \in R^{m+1}$, and scalars $c_P, c_L \in R$ satisfy

$$p_J(P) = c_P \det(K_P)_J \quad (|J| = t_1),$$

$$L\beta = 0, \quad p_{\{\ell\}}(L) = c_L \beta_\ell \quad (0 \leq \ell \leq m).$$

Let $\lambda \in R^N$ be a lift with $S\lambda = \beta$, extend K_P by zero columns to $K_0 = (K_P \ 0)$, and order

$$K = \begin{pmatrix} K_0 \\ \lambda^\top \end{pmatrix}$$

by the t_1 inherited rows followed by the lifted outer row. Then for every $(t_1 + 1)$ -element column set I ,

$$\boxed{p_I(M) = (-1)^{t_1} c_P c_L \det K_I.}$$

Proof. The block matrix S inherits the complementary-minor identity

$$p_J(S) = c_P \det(K_0)_J \quad (|J| = t_1). \tag{3.1}$$

Indeed, if J meets the identity block, both sides vanish by block rank; if J lies in the P -source, the complementary square matrix has diagonal blocks $P_{\hat{J}}$ and I_q .

Fix $I = \{i_0 < \dots < i_{t_1}\}$ and put $C = I^c$, so $|C| = m$. Cauchy – Binet, with the intermediate rows in increasing order, gives

$$\det M_C = \sum_{\ell=0}^m \det L_{\ell} \det S_{\ell,C} = c_L \sum_{\ell=0}^m (-1)^{\ell} \beta_{\ell} \det S_{\ell,C}. \quad (3.2)$$

Expanding the $(m+1)$ -square matrix $[S_C \mid \beta]$ along its final column yields

$$\sum_{\ell=0}^m (-1)^{\ell} \beta_{\ell} \det S_{\ell,C} = (-1)^m \det[S_C \mid \beta]. \quad (3.3)$$

Since $\beta = S\lambda$, multilinearity in the final column gives

$$\det[S_C \mid \beta] = \sum_{a=0}^{t_1} \lambda_{i_a} \det[S_C \mid S_{i_a}]. \quad (3.4)$$

There are $m - i_a + a$ elements of C larger than i_a . Move the last column S_{i_a} into increasing position and apply (3.1):

$$\begin{aligned} \det[S_C \mid S_{i_a}] &= (-1)^{m-i_a+a} \det S_{C \cup \{i_a\}} \\ &= (-1)^{m+a+\sum I_{C_P}} \det(K_0)_{I \setminus \{i_a\}}. \end{aligned} \quad (3.5)$$

The two occurrences of i_a cancel modulo two. On the other hand, expansion of $\det K_I$ along its last row is

$$\det K_I = \sum_{a=0}^{t_1} (-1)^{t_1+a} \lambda_{i_a} \det(K_0)_{I \setminus \{i_a\}}. \quad (3.6)$$

Equations (3.4) – (3.6) imply

$$\det[S_C \mid \beta] = (-1)^{m+\sum I+t_1} c_P \det K_I. \quad (3.7)$$

Substitution into (3.2) – (3.3) gives $\det M_C = (-1)^{\sum I+t_1} c_P c_L \det K_I$. Multiplying by the defining factor $(-1)^{\sum I}$ proves the result. $\square$

Remark 3.2. The factor $(-1)^{t_1}$ depends on the fixed block order. It cannot be recovered from the unordered multiset of degrees. This is visible already for the ordered triples $(1, 1, 2)$, $(1, 2, 1)$, and $(2, 1, 1)$, whose final orientations are respectively negative, positive, and negative.

4. Affine Plücker lift of the multiplication Jacobian

Define

$$E(\mathbf{d}) = \sum_{1 \leq i < j \leq k} d_i(d_j + 1) + \binom{k+1}{2} + 1, \quad \varepsilon_{\mathbf{d}} = (-1)^{E(\mathbf{d})}.$$

Since

$$\binom{k+1}{2} + 1 - \binom{k-1}{2} = 2k,$$

the same orientation is obtained by replacing the final two terms in $E(\mathbf{d})$ with $\binom{k-1}{2}$. This parity-equivalent form makes the inductive contribution of the $k-2$ inherited kernel directions transparent.

Theorem 4.1 (affine Plücker lift of the classical multiplication Jacobian). For every $k \geq 2$, every positive ordered degree vector $\mathbf{d}$, and every $(k-1)$ -element source-column set I ,

$$\boxed{p_I(Dm_{\mathbf{d}}) = \varepsilon_{\mathbf{d}} \Delta_{\mathbf{d}} \det K_I.}$$

The equality is an identity in the universal coefficient ring over $\mathbb{Z}$, and hence is compatible with arbitrary base change.

The square-chart scalar and the pairwise-coprime regularity criterion are classical antecedents. The content recorded by Theorem 4.1 is their simultaneous non-monic affine lift: it specifies every complementary minor, the matching torus Plücker coordinate, and the orientation in one universal integral identity. This exact packaging is used as the structural bridge to the character-lattice calculation; it is not presented as a new resultant determinant in isolation.

Proof. We induct on k . For $k = 2$, the self-contained two-factor argument in Appendix E gives

$$p_{\{i\}}(Dm_{d_1, d_2}) = (-1)^{d_1(d_2+1)} \text{Res}(A_1, A_2) K_i.$$

The proposed exponent differs from $d_1(d_2+1)$ by $\binom{3}{2} + 1 = 4$, so the signs agree.

Assume the theorem for $k-1$ factors. Put

$$D' = d_1 + \cdots + d_{k-1}, \quad B = A_1 \cdots A_{k-1}.$$

Factor multiplication as

$$\prod_{i=1}^k V_{d_i} \xrightarrow{m_{d_1, \dots, d_{k-1}} \times \text{id}} V_{D'} \times V_{d_k} \xrightarrow{m_{D', d_k}} V_D.$$

The inner Plücker scalar is

$$\varepsilon_{d_1, \dots, d_{k-1}} \prod_{1 \leq i < j < k} R_{ij}.$$

The outer two-factor scalar is

$$(-1)^{D'(d_k+1)} \text{Res}(B, A_k).$$

Resultant multiplicativity in the fixed Sylvester convention gives the exact identity

$$\text{Res}(B, A_k) = \prod_{i < k} R_{ik}.$$

No permutation of factor blocks is made, so no additional resultant sign appears.

For the first $k-1$ factors, use the inherited kernel rows

$$\eta_b = (0, \dots, A_b, \dots, -A_{k-1}, 0), \quad b = 1, \dots, k-2.$$

A lift of the outer kernel $(B, -A_k)$ is

$$\ell = (0, \dots, 0, A_{k-1}, -A_k),$$

because differentiating the inner product in the A_{k-1} scaling direction sends A_{k-1} to B . The standard torus rows satisfy

$$\kappa_b = \eta_b + \ell \quad (1 \leq b \leq k-2), \quad \kappa_{k-1} = \ell.$$

The corresponding row-basis transformation is unitriangular and has determinant one. Lemma 3.1, with $t_1 = k-2$, therefore gives the parity recurrence

$$e_k \equiv e_{k-1} + D'(d_k + 1) + (k-2) \pmod{2}.$$

The closed exponent satisfies

$$E(d_1, \dots, d_k) - E(d_1, \dots, d_{k-1}) = D'(d_k + 1) + k,$$

and $k \equiv k-2 \pmod{2}$. The scalar resultants combine to Δ_d , while the determinant-one row change gives the displayed standard kernel minors. This completes the induction. $\square$

For three factors the sign can be written more compactly as

$$\varepsilon_{d_1, d_2, d_3} = (-1)^{d_1(d_2+1) + (d_1+d_2)(d_3+1)+1}.$$

Thus every two-column deletion satisfies

$$(-1)^{i+j} \det M_{\widehat{\{i,j\}}} = \varepsilon_{d_1, d_2, d_3} R_{12} R_{13} R_{23} \det K_{\{i,j\}}.$$

This is the first higher-rank case of the induction used in this paper.

4.1 Classical rank criterion and scheme-theoretic refinement

The coordinate identities yield the exact ideal equality

$$\boxed{I_{D+1}(Dm_d) = \Delta_d I_{k-1}(K).}$$

On the locus where every factor is nonzero, K has row rank $k-1$. On the pairwise-coprime locus, these rows are the entire kernel, consistently with the known monic multifactor rank criterion (Chaperon and López de Medrano 2009, Theorem 3). To see this in the present affine coordinates, let a tangent vector $(H_1, \dots, H_k)$ satisfy

$$\sum_{i=1}^k H_i \prod_{j \neq i} A_j = 0.$$

Reduce the equation modulo A_i . Since A_i is coprime to $\prod_{j \neq i} A_j$, it divides H_i . Degree equality gives $H_i = c_i A_i$ for a scalar c_i . Substitution gives $\sum_i c_i = 0$, precisely the tangent equation of the product-one torus.

The classical criterion identifies pairwise root collision as the generic rank-loss set. The ideal identity refines it in the universal affine coefficient ring by retaining every maximal minor. Each generic pairwise

resultant is a primitive irreducible polynomial and the different R_{ij} are nonassociate (Jouanolou 1991). At the height-one prime (R_{ij}) , the K -minor obtained by choosing one fixed coefficient column from each of the first $k - 1$ factor blocks is a product of those coefficients and is not divisible by R_{ij} ; it is therefore a unit in the height-one localisation. All other resultants are units there, so the maximal-minor ideal has valuation one.

The exact statement after base change is slightly different. Let $A_{\mathbb{Z}}$ be the universal coefficient ring, let $A_{\mathbb{Z}} \rightarrow B$ be any ring map, and put $J_B = I_{k-1}(K_B)$. Polynomial base change gives unconditionally

$$I_{D+1}(Dm_{d,B}) = \Delta_{d,B} J_B. \quad (4.4)$$

If B is a Noetherian normal domain and $\mathfrak{p}$ is height one, then $B_{\mathfrak{p}}$ is a DVR and

$$v_{\mathfrak{p}}(I_{D+1}(Dm_{d,B})) = \sum_{e \in E(K_k)} v_{\mathfrak{p}}(R_{e,B}) + v_{\mathfrak{p}}(J_B), \quad (4.5)$$

where the valuation of an ideal is the minimum valuation of its generators. Consequently the order is exactly one along $\mathfrak{p}$ provided that

1. $v_{\mathfrak{p}}(R_{ij,B}) = 1$;
2. every $R_{e,B}$ with $e \neq ij$ is a unit at $\mathfrak{p}$; and
3. some maximal K_B -minor is a unit there.

Equivalently, if a prime effective Cartier divisor has local equation s , the same conclusion holds when the pulled-back $R_{ij} = us$ for a unit u , all other resultants and some maximal K -minor are units at its generic point. A ramified pullback $R_{ij} = us^m$ contributes order m , even when the underlying component is reduced. Second collisions and torus-rank defects contribute the additional terms visible in (4.5). Intersections and zero-factor strata can therefore carry extra scheme structure through J_B . The theorem does not assert that the determinantal ideal is globally principal on the affine source.

Remark 4.2. The first power of each pairwise resultant also follows from multidegree bookkeeping once divisibility and the kernel tensor are known. The induction is stronger: it fixes the complete vector and its orientation without an irreducibility or generic-divisor argument.

5. Multi-border contraction

Let $g_1, \dots, g_t$ be polynomials in the source coefficients, and let G be their $t \times N$ gradient matrix,

$$G_{aj} = \frac{\partial g_a}{\partial x_j}.$$

The torus vector fields associated with the rows of K are

$$\delta_b = \sum_{j=0}^{N-1} K_{bj} \frac{\partial}{\partial x_j}, \quad b = 1, \dots, t.$$

Thus $(KG^T)_{ba} = \delta_b g_a$.

Corollary 5.1 (multi-border contraction). With the multiplication rows followed by the gradient rows in the order $g_1, \dots, g_t$,

$$\det \begin{pmatrix} M \\ G \end{pmatrix} = (-1)^{nt + \binom{t}{2}} \varepsilon_d \Delta_d \det(\delta_b g_a)_{a,b=1}^t.$$

Proof. Expand the determinant along the first n rows. If I is the set of t columns assigned to the gradient block, the corresponding term is the product of $\det M_{\hat{I}}$ and $\det G_I$, times the Laplace sign. Substitution of $p_I(M) = (-1)^{\sum I} \det M_{\hat{I}}$ cancels the part of the Laplace sign depending on I . The remaining parity is $nt + \binom{t}{2}$. Theorem 4.1 then gives

$$(-1)^{nt + \binom{t}{2}} \varepsilon_d \Delta_d \sum_{|I|=t} \det K_I \det G_I.$$

Cauchy – Binet identifies the sum with $\det(KG^T)$. Transposing this final $t \times t$ matrix does not change its determinant, and its entries are the displayed vertical derivatives. $\square$

Suppose now that each g_a is a T -semi-invariant. Relative to the cocharacter basis $e_b - e_k$, let its integer character be the a -th row of W . Then

$$\delta_b g_a = W_{ab} g_a,$$

so the vertical matrix is $\text{diag}(g_1, \dots, g_t)W$.

Corollary 5.2 (character determinant).

$$\det D(m_d, g_1, \dots, g_t) = (-1)^{nt + \binom{t}{2}} \varepsilon_d \Delta_d \det(W) \prod_{a=1}^t g_a.$$

For $k = 2$, W has one entry. If g has bidegree (p, q) , that entry is $p - q$. The degree-difference theorem is therefore exactly the rank-one case of the character determinant.

Corollary 5.2 separates two obstructions. The collision factor Δ records additional kernel directions caused by common roots. The integer $\det W$ records whether the chosen borders recover the generic torus directions. Even when $\det W \neq 0$, the collision divisor remains.

6. Resultant-character minors via weighted signless incidence

6.1 Edge characters

Under factor scaling, the pairwise resultant transforms as

$$R_{ij}(\lambda_i A_i, \lambda_j A_j) = \lambda_i^{d_j} \lambda_j^{d_i} R_{ij}(A_i, A_j).$$

On the cocharacter lattice

$$X_*(T) = \{u = (u_1, \dots, u_k) \in \mathbb{Z}^k : \sum_i u_i = 0\},$$

the edge ij therefore defines

$$\chi_{ij}(u) = d_j u_i + d_i u_j.$$

Use coordinates $u_1, \dots, u_{k-1}$ and eliminate $u_k = -\sum_{b < k} u_b$. For a set H of $k - 1$ edges, let W_H be the resulting square character matrix.

The graph associated with H need not be connected. Let $\deg_H(i)$ denote the valence of vertex i . A connected component is odd unicyclic if its number of edges equals its number of vertices and its unique cycle has odd length.

6.2 Signed-graphic reduction and graph minors

Once the edge characters are written in these coordinates, their maximal-minor calculation is a degree-weighted specialization of classical signless-incidence minor theory. Up to transpose and edge-orientation conventions, the relevant matrix is the signed incidence matrix of the all-negative signed graph $-H$. In that language, balance is bipartiteness, the rank defect counts bipartite components, and an unbalanced unicyclic component contributes a factor two. These support, rank, and power-of-two facts are classical (Zaslavsky 1982, sec. 7D, Corollaries 7D.1 and 7D.3(d),(j), and sec. 8A, Lemmas 8A.2 – 8A.3; Zaslavsky 1983). We do not use the uncorrected wording of Corollary 7D.3(g). The theorem below records the factorisation-specific cross-degree transformation and the evaluation of the appended degree row; it makes no claim to a new signed-incidence classification.

Theorem 6.1 (resultant-character minor formula). The determinant $\det W_H$ can be nonzero only when H has exactly one tree component and every other component is odd unicyclic. Suppose this condition holds, let c be the number of components, and let $P \sqcup Q$ be the bipartition of the tree component. Then

$$|\det W_H| = 2^{c-1} \left| \sum_{i \in P} d_i - \sum_{j \in Q} d_j \right| \prod_{i=1}^k d_i^{\deg_H(i)-1}.$$

The right-hand side is integral. When the tree component is an isolated vertex, use the bipartition $P = \{i\}, Q = \emptyset$; its degree balance d_i cancels the apparent exponent -1 . Within the stated component pattern, the determinant is nonzero exactly when the tree component has nonzero bipartite degree balance.

Proof. Before eliminating u_k , form the $(k-1) \times k$ matrix C_H whose edge ij row has entry d_j in column i , entry d_i in column j , and zero elsewhere. Append a last row $(1, \dots, 1)$. Let $S \in \text{GL}_k(\mathbb{Z})$ have columns $e_1 - e_k, \dots, e_{k-1} - e_k, e_k$. Its determinant is one, and

$$\begin{pmatrix} C_H \\ 1^\top \end{pmatrix} S = \begin{pmatrix} W_H & * \\ 0 & 1 \end{pmatrix}.$$

Thus the absolute determinant of the augmented matrix is exactly $|\det W_H|$; in particular, eliminating the torus constraint introduces no hidden factor of k .

Multiply column i by d_i . The edge ij row now has the common factor $d_i d_j$, with a one in each endpoint column after that factor is removed. The last row becomes $(d_1, \dots, d_k)$. Comparing determinants before cancelling any factor gives the integral identity

$$\left(\prod_i d_i \right) |\det W_H| = \left(\prod_i d_i^{\deg_H(i)} \right) \left| \det \begin{pmatrix} B_H & \\ d_1 & \dots & d_k \end{pmatrix} \right|. \quad (6.1)$$

If every vertex is incident to an edge, cancellation gives

$$|\det W_H| = \left(\prod_{i=1}^k d_i^{\deg_H(i)-1} \right) \left| \det \begin{pmatrix} B_H & \\ d_1 & \dots & d_k \end{pmatrix} \right|,$$

where B_H is the unoriented edge – vertex incidence matrix. If the unique tree component is an isolated vertex i , the appended determinant in (6.1) contains the factor d_i ; cancelling that factor first gives the same displayed formula with no negative exponent. Thus no division by a degree is used to establish integrality.

Reorder rows and columns by connected component. A tree component on v vertices supplies $v-1$ independent incidence rows. Appending the restricted degree row produces determinant, up to sign,

$$\sum_{i \in P} d_i - \sum_{j \in Q} d_j,$$

because the signed bipartition vector spans the kernel of its incidence rows. An odd-unicyclic component has a square incidence matrix of absolute determinant two. An even-unicyclic component is singular. A component with more than one independent cycle produces row dependence in a square maximal minor, while two tree components leave more than one missing incidence row for the single degree border. Since H has $k - 1$ edges, nonzero determinant therefore requires one tree component and only odd-unicyclic remaining components. Multiplying the component determinants proves the formula. $\square$

The incidence facts used in the last paragraph are classical. Grossman, Kulkarni, and Schochetman determine the minors and Smith normal form of unoriented graph incidence matrices (Grossman et al. 1995). The odd-unicyclic invertibility criterion also appears in later inverse formulas (Hessert and Mallik 2023). The factorisation-specific content is the weighted transformation from resultant characters to that incidence matrix.

Corollary 6.2 (tree formula). If $H = T$ is a spanning tree, then

$$|\det W_T| = \left| \sum_{i \in P} d_i - \sum_{j \in Q} d_j \right| \prod_i d_i^{\deg_T(i)-1}.$$

For a star centred at vertex k ,

$$|\det W_T| = d_k^{k-2} \left| d_k - \sum_{i < k} d_i \right|.$$

For three equal linear factors, every star has determinant one. Equal degrees therefore do not force degeneration when the torus rank is greater than one. This contrasts with the two-factor resultant character, whose single weight is the degree difference.

6.3 A three-linear-factor example

Let $d_1 = d_2 = d_3 = 1$ and write $A_i = x_i X + y_i Y$. If

$$A_j A_\ell = p_i X^2 + q_i XY + r_i Y^2, \quad \{i, j, \ell\} = \{1, 2, 3\},$$

then, in block-major source order $(x_1, y_1, x_2, y_2, x_3, y_3)$, the multiplication differential and ordered torus kernel are

$$M = \begin{pmatrix} p_1 & 0 & p_2 & 0 & p_3 & 0 \\ q_1 & p_1 & q_2 & p_2 & q_3 & p_3 \\ r_1 & q_1 & r_2 & q_2 & r_3 & q_3 \\ 0 & r_1 & 0 & r_2 & 0 & r_3 \end{pmatrix}, \quad K = \begin{pmatrix} x_1 & y_1 & 0 & 0 & -x_3 & -y_3 \\ 0 & 0 & x_2 & y_2 & -x_3 & -y_3 \end{pmatrix}.$$

Here $\varepsilon_{(1,1,1)} = -1$. For example, deleting the x_1 - and x_2 -columns gives

$$p_{\{0,2\}}(M) = -R_{12}R_{13}R_{23} \det \begin{pmatrix} x_1 & 0 \\ 0 & x_2 \end{pmatrix} = -x_1 x_2 R_{12}R_{13}R_{23}.$$

Choose the path normalising functions (R_{12}, R_{23}) . In the cocharacter basis $(e_1 - e_3, e_2 - e_3)$, their character matrix is

$$W_{\{12,23\}} = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \quad \det W_{\{12,23\}} = 1.$$

The full edge-character matrix is

$$W_E = \begin{pmatrix} 1 & 1 \\ 0 & -1 \\ -1 & 0 \end{pmatrix},$$

which has Smith form $\text{diag}(1, 1)$. Thus two resultant normalising functions remove the two-dimensional scaling torus without a residual finite group scheme. The collision factor $R_{12}R_{13}R_{23}$ nevertheless remains in the bordered Jacobian. Replacing the degrees by $(1, 1, 2)$ changes the same path matrix to $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, of determinant two; the corresponding residual group scheme is μ_2 , nonreduced in characteristic two.

7. Exact Smith invariants of the unit-character lattice

7.1 Regular units and arithmetic-matroid multiplicity

Let $A_{\mathbb{Z}}$ be the universal coefficient ring and put

$$U = D(\Delta_d) \subset \text{Spec } A_{\mathbb{Z}}.$$

The pairwise resultants are primitive irreducibles and are pairwise nonassociate in the UFD $A_{\mathbb{Z}}$. Hence

$$\Gamma(U, \mathcal{O}_U)^\times = \left\{ \varepsilon \prod_{i < j} R_{ij}^{n_{ij}} : \varepsilon \in \{\pm 1\}, n_{ij} \in \mathbb{Z} \right\}. \quad (7.1)$$

Indeed, if f/Δ^N and g/Δ^M are inverse, then $fg = \Delta^{N+M}$, and unique factorisation forces every irreducible factor of f and g to be one of the R_{ij} . Valuations at the distinct height-one primes (R_{ij}) give uniqueness. Over a field, the same proof replaces $\{\pm 1\}$ by the field's multiplicative group.

Thus all regular units on U are semi-invariant Laurent resultant monomials. Their characters form exactly the row lattice L of the complete edge-character matrix W_E . This is the intrinsic character lattice of regular units on the fixed pairwise-coprime open U . It is not the lattice of every polynomial semi-invariant, and it can enlarge after deleting further divisors. We call

$$D(X^*(T)/L)$$

the **unit-character residual group** of U .

Set

$$g = \gcd(d_1, \dots, d_k), \quad e_i = d_i/g, \quad M = X^*(T) = \mathbb{Z}^k/\mathbb{Z}1.$$

For the primitive vector $\mathbf{e}$, let

$$\chi_{ij} = [e_j \varepsilon_i + e_i \varepsilon_j] \in M, \quad C_{\mathbf{e}} = M/\langle \chi_{ij} : i < j \rangle,$$

and define

$$h(\mathbf{e}) = \gcd_{\substack{H \subset E(K_k) \\ |H|=k-1}} |\det W_H(\mathbf{e})|. \quad (7.2)$$

This gcd is standard representable arithmetic-matroid data. For a sublist S of the primitive edge characters, put

$$m(S) = [M \cap \text{span}_{\mathbb{R}}(S) : \langle S \rangle_{\mathbb{Z}}].$$

Then $h(\mathbf{e}) = m(E)$, while a full-rank square sublist has $m(H) = |\det W_H|$. The equality between $m(E)$ and the gcd of basis multiplicities is the standard GCD rule (Moci 2012, sec. 2.2; D’Adderio and Moci 2013, secs. 1.4 – 1.5). For the original list $\mathbf{d} = g\mathbf{e}$, the full-list multiplicity is $g^{k-1}h(\mathbf{e})$, not $h(\mathbf{e})$.

7.2 A one-generator local presentation

Write $x_i \in M$ for the image of ε_i , so that $\sum_i x_i = 0$. The edge relations are

$$q_{ij} = e_j x_i + e_i x_j.$$

For a prime p , let $R_p = \mathbb{Z}_{(p)}$ and use the convention $v_p(0) = \infty$.

Theorem 7.1 (exact local Smith presentation). Let $k \geq 3$, let $\mathbf{e}$ be primitive, and choose an index a with $p \nmid e_a$. Put

$$D_a = e_a - \sum_{i \neq a} e_i$$

and

$$I_{p,a} = (D_a, 2e_i e_j : i < j, i, j \neq a) R_p.$$

Then $1 \mapsto x_a$ induces an isomorphism

$$\boxed{C_{\mathbf{e}} \otimes R_p \cong R_p / I_{p,a}.} \quad (7.3)$$

In particular, the presentation is valid for $p = 2$, for $k = 3$, and when $D_a = 0$.

Proof. Primitivity supplies a p -unit e_a . Over R_p , each star relation

$$e_i x_a + e_a x_i = 0$$

eliminates $x_i = -(e_i/e_a)x_a$. The ambient relation becomes $(D_a/e_a)x_a = 0$, which is equivalent to $D_a x_a = 0$. Each remaining edge ij , with $i, j \neq a$, becomes

$$-\frac{2e_i e_j}{e_a} x_a = 0,$$

equivalent to $2e_i e_j x_a = 0$. These are all generators and relations, so the Tietze elimination proves the module isomorphism, not merely an equality of orders. No division by 2, D_a , or a nonunit degree occurs. $\square$

The ideal in (7.3) is independent of the admissible pivot: two such presentations are the annihilator of the same canonical local module, and the edge between the pivots changes the displayed cyclic generator by a unit.

Corollary 7.2 (all local valuations). For any p -unit pivot a ,

$$v_p(h(\mathbf{e})) = m_p(\mathbf{e}) := \min \left(v_p(D_a), \min_{\substack{i < j \\ i, j \neq a}} [v_p(2) + v_p(e_i) + v_p(e_j)] \right). \quad (7.4)$$

Moreover,

$$C_{\mathbf{e}} \otimes R_p \cong R_p / (p^{m_p(\mathbf{e})}), \quad C_{\mathbf{e}} \cong \mathbb{Z} / h(\mathbf{e})\mathbb{Z}. \quad (7.5)$$

Proof. Every ideal in the DVR R_p is generated by an element of least valuation, which gives the first isomorphism. The zeroth Fitting ideal of the local cokernel is both $I_{p,a}$ and the localisation of the ideal generated by all maximal minors. Its positive generator over $\mathbb{Z}$ is $h(\mathbf{e})$, proving (7.4). The local presentation is torsion, hence $C_{\mathbf{e}} \otimes \mathbb{Q} = 0$; the finitely generated abelian group $C_{\mathbf{e}}$ is therefore finite. Its primary localisations are cyclic, and their direct sum has pairwise coprime orders, so the global group is cyclic of order h . $\square$

This quotient module is the full-set module of the represented matroid over $\mathbb{Z}$. Arithmetic multiplicity remembers only its order; the matroid-over-a-ring and DVR frameworks retain its isomorphism type and local invariant sequence (Fink and Moci 2016, Definition 2.1, eq. (2.1), secs. 5 and 6.1). Theorem 7.1 is an explicit computation of that standard object for this cross-weighted factorisation list.

7.3 Closed formula and tree-gcd equality

For an odd prime p , (7.4) is nonzero precisely when exactly two degrees, say e_a, e_b , are p -units and $e_a \equiv e_b \pmod{p}$. In that case

$$m_p(\mathbf{e}) = \min \left(v_p(e_a - e_b), \min_{c \notin \{a,b\}} v_p(e_c) \right). \quad (7.6)$$

For $p = 2$, let s be the number of odd e_i . Primitivity gives $s \geq 1$, and

$$m_2(\mathbf{e}) = \begin{cases} 0, & s \text{ odd,} \\ 1, & s \geq 4 \text{ and } s \text{ even,} \\ \min(v_2(D_a), 1 + \min_{c \notin \{a,b\}} v_2(e_c)), & s = 2, \end{cases} \quad (7.7)$$

where a, b are the two odd indices in the last line. Swapping a and b does not change the truncated valuation: if $E = \sum_{c \notin \{a,b\}} e_c$ and $q = \min_c v_2(e_c)$, then $D_a + D_b = -2E$ is divisible by 2^{q+1} .

The primewise description assembles without factoring a maximal minor. Define

$$G_{ab} = \gcd\{e_c : c \notin \{a, b\}\}, \quad Q_{ab} = \gcd(|e_a - e_b|, G_{ab}),$$

and let $Q_{ab}^{\text{odd}} = Q_{ab} / 2^{v_2(Q_{ab})}$. Put

$$\eta_2(\mathbf{e}) = \begin{cases} 0, & s \text{ odd,} \\ 1, & s \geq 4 \text{ and } s \text{ even,} \\ \min(v_2(D_a), 1 + v_2(G_{ab})), & s = 2. \end{cases}$$

Theorem 7.3 (closed global index). One has

$$h(\mathbf{e}) = 2^{\eta_2(\mathbf{e})} \prod_{1 \leq a < b \leq k} Q_{ab}^{\text{odd}}. \quad (7.8)$$

The odd factors in the product are pairwise coprime. Consequently $h(\mathbf{e})$ is computable with $O(k^2)$ integer gcd operations. For each fixed a , prefix and suffix gcds of the other $k - 1$ degrees compute every excluded-pair gcd G_{ab} in $O(k)$ gcd operations; repeating over a gives the stated bound. This is an arithmetic-operation count, not a unit-cost bit-complexity claim.

Proof. If an odd prime divides Q_{ab} , it divides every degree outside $\{a, b\}$ and divides $e_a - e_b$. It cannot divide e_a or e_b , else it would divide every primitive degree. Thus $\{a, b\}$ is exactly the pair of unit indices, and (7.6) gives $v_p(h) = v_p(Q_{ab})$. Conversely every exceptional odd prime divides that Q_{ab} . The unit pair is unique, so no odd prime divides two different Q_{ab} . Formula (7.7) supplies the remaining exponent at two. Equality follows prime by prime. $\square$

The full-list gcd is even more economical than its definition suggests.

Theorem 7.4 (spanning trees determine the full index). If

$$\tau(\mathbf{e}) = \gcd_{T \text{ spanning tree of } K_k} |\det W_T(\mathbf{e})|,$$

then $\tau(\mathbf{e}) = h(\mathbf{e})$. After localising at p , choose any p -unit pivot a . The star centred at a and its associated $\binom{k-1}{2}$ bent stars generate the full maximal-minor ideal.

Proof. By the spanning-tree determinant formula in Corollary 6.2, after choosing a p -unit vertex a , the star centred at a has determinant, up to sign,

$$e_a^{k-2} D_a,$$

so it generates D_a up to a unit in R_p . For distinct $i, j \neq a$, remove aj from the star and add ij . The resulting bent star has determinant

$$\pm e_a^{k-3} e_i (D_a + 2e_j).$$

Subtracting $e_i D_a$ yields $2e_i e_j$. Hence these tree minors contain the ideal $I_{p,a}$. Conversely every tree determinant is a maximal minor, and the ideal of all maximal minors is $\text{Fitt}_0(C_{\mathbf{e}} \otimes R_p) = I_{p,a}$. The two ideals therefore agree at every prime, so their positive integer gcds agree. $\square$

Odd-unicyclic bases remain relevant as individual arithmetic-matroid multiplicities and square-chart determinants. Theorem 7.4 says only that they do not lower the single full-list saturation index.

7.4 Nonprimitive Smith form and residual group scheme

Because $W_E(\mathbf{d}) = gW_E(\mathbf{e})$, the primitive calculation gives the complete answer.

Corollary 7.5 (complete Smith form). For $k \geq 3$,

$$\text{SNF}(W_E(\mathbf{d})) = \text{diag}(\underbrace{g, \dots, g}_{k-2}, gh(\mathbf{e})). \quad (7.9)$$

Thus

$$X^*(T)/L \cong (\mathbb{Z}/g\mathbb{Z})^{k-2} \oplus \mathbb{Z}/(gh(\mathbf{e}))\mathbb{Z}. \quad (7.10)$$

Indeed, the unimodular matrices that diagonalise the primitive matrix put its scalar multiple into (7.9), which already satisfies Smith divisibility.

Over $\text{Spec } \mathbb{Z}$, the unit-character residual group is noncanonically

$$\boxed{D(X^*(T)/L) \cong \mu_g^{k-2} \times \mu_{gh(\mathbf{e})}} \quad (7.11)$$

It is finite locally free of rank $g^{k-1}h(\mathbf{e})$. Over a field of characteristic $p > 0$, write $g = p^\gamma g'$, $h = p^m h'$, with $p \nmid g' h'$. Its connected diagonalizable factor and prime-to- p étale direct factor are

$$\mu_{p^\gamma}^{k-2} \times \mu_{p^{\gamma+m}}, \quad \mu_{g'}^{k-2} \times \mu_{g' h'}, \quad (7.12)$$

respectively. Their ranks are $p^{\gamma(k-1)+m}$ and $(g')^{k-1}h'$. Thus the scheme is étale exactly when $p \nmid gh$, and the number of geometric points in an algebraic closure is $(g')^{k-1}h'$, not its total scheme rank in bad characteristic.

For comparison with toric arrangements, over $\mathbb{C}$ the standard identity $m(S) = \#\pi_0(\bigcap_{\chi \in S} \ker \chi)$ gives exactly the same full-list order (Moci 2012, Lemma 5.4; D'Adderio and Moci 2013, Lemma 4.1). The group-scheme statement (7.11) is the characteristic-sensitive refinement.

The rectangular all-edge character morphism

$$T \longrightarrow (\mathbb{G}_m)^{E(K_k)}$$

has scheme-theoretic image $D(L)$. The induced map $T \rightarrow D(L)$ is finite locally free of degree $[X^*(T) : L] = g^{k-1}h$, with kernel (7.11). On coordinate rings this is the free inclusion $\mathbb{Z}[L] \subset \mathbb{Z}[X^*(T)]$, with a basis given by coset representatives. The degree is the degree onto this image, not a claim about an unlabelled rectangular target containing every root-partition branch.

7.5 Diagnostic examples

Degrees	$h(\mathbf{e})$	Smith form	Diagnostic point
(1, 1, 1)	1	diag(1, 1)	Saturated full list
(1, 1, 3)	3	diag(1, 3)	Odd bad prime 3
(1, 1, 9)	9	diag(1, 9)	$v_3(h) = 2$
(1, 1, 4)	4	diag(1, 4)	$v_2(h) = 2$
(1, 5, 4)	8	diag(1, 8)	$v_2(h) = 3$
(1, 2, 2)	1	diag(1, 1)	Full list saturated; square edge minors have absolute values 3, 2, 2
(1, 1, 1, 1)	2	diag(1, 1, 2)	Four odd entries give exactly one factor of 2
(2, 2, 18) = 2(1, 1, 9)	9	diag(2, 18)	Nonprimitive content and odd valuation

The families

$$h(1, 1, N) = N, \quad h(1, 1 + 2^q, 2^q) = 2^{q+1} \quad (q \geq 1) \quad (7.13)$$

realise arbitrary higher odd and two-adic valuations. The vector $(1, 2, 2)$ separates full-list saturation from square edge charts: no two individual resultant characters form a unimodular basis, despite $h = 1$.

8. Generic-degree synthesis and étaleness

8.1 Square semi-invariant charts

Let $\mathbb{k}$ be an algebraically closed field of characteristic $p \geq 0$. A polynomial g is a T -semi-invariant of character χ if $g(\lambda \cdot A) = \chi(\lambda)g(A)$. Suppose $g_1, \dots, g_t$ are nonzero semi-invariants whose character matrix $W \in M_t(\mathbb{Z})$ has nonzero determinant. Set

$$\Phi = (m_d, g_1, \dots, g_t) : \prod_i V_{d_i} \longrightarrow V_D \times \mathbb{A}^t$$

and

$$X^\circ = D \left(\Delta_d \prod_{a=1}^t g_a \right).$$

Here the generic scheme degree means the degree of the induced finite extension of function fields. The proof below constructs a dense target open on which the map is finite locally free, so this degree is also the constant scheme length of every fibre over that open.

The following statement combines the classical labelled root-partition degree with the scheme order of the torus isogeny defined by W . Its role is to translate the character-lattice data into the geometry of a normalised factorisation chart.

Theorem 8.1 (finite normalised factorisation cover). Fix the labelled degrees, let $\mathbb{k}$ be algebraically closed, and let $g_1, \dots, g_t$ be nonzero semi-invariants with a square character matrix W of nonzero determinant. There is a dense open $V_D^{\text{good}} \subset V_D$ such that the restriction of Φ over $V_D^{\text{good}} \times (\mathbb{G}_m)^t$ is finite locally free. In particular, Φ is dominant and generically finite, with

$$\deg_{\text{gen}} \Phi = |\det W| \frac{D!}{\prod_i d_i!}.$$

If the Smith form of W is $\text{diag}(s_1, \dots, s_t)$, then the torus part of each generic fibre is a translate of

$$\ker \chi_W \cong \prod_{i=1}^t \mu_{s_i}.$$

It has scheme order $|\det W|$. If $p > 0$, write $s_i = p^{a_i} s'_i$ with $p \nmid s'_i$. The generic separable degree is

$$\left(\prod_i s'_i \right) \frac{D!}{\prod_i d_i!},$$

the number of geometric points in an algebraic closure of a generic fibre is the same number, and the inseparable factor is $p^{\sum_i a_i} = p^{v_p(\det W)}$. On X° , the map is étale exactly when $p \nmid \det W$, with the characteristic-zero convention.

Proof. Let V_D^{sf} be the open set of nonzero binary forms with D distinct geometric roots, and define

$$Z^{\text{sf}} = V_D^{\text{sf}} \times_{\mathbb{P}(V_D)} \prod_{i=1}^k \mathbb{P}(V_{d_i}),$$

where the second map is projective multiplication and the projective source is restricted to the inverse image of the squarefree locus. The projection $\pi : Z^{\text{sf}} \rightarrow V_D^{\text{sf}}$ is the finite étale root-partition cover: a point above $F \in V_D^{\text{sf}}$ is a projective factorisation obtained by partitioning the roots of F into labelled blocks of sizes $(d_1, \dots, d_k)$. Its degree is

$$\nu = \frac{D!}{\prod_i d_i!}. \quad (8.1)$$

This is also the standard projective multiplication-fibre description (Breiding et al. 2024, sec. 10.2, Proposition 10.7 and Corollary 10.8, p. 130). Finite étaleness follows from the free symmetric-group action on ordered distinct-root configurations; equivalently, projective multiplication is finite and its differential is an isomorphism here because the affine differential has only the T -tangent kernel (Chaparon and López de Medrano 2009, Theorem 3). Scheme-theoretically, the free actions of the relevant constant symmetric groups have scheme quotients and give fppf torsors by (The Stacks Project Authors 2026, tag 07S7); after base change along the torsor they are disjoint unions of copies of the base. Finite locally free degree and étaleness descend fpqc-locally (The Stacks Project Authors 2026, tags 02VO and 02VN).

Put $U^{\text{sf}} = m_d^{-1}(V_D^{\text{sf}})$. The map

$$q : U^{\text{sf}} \longrightarrow Z^{\text{sf}}, \quad (A_i)_i \mapsto \left(\prod_i A_i, ([A_i])_i \right),$$

is a T -torsor: two affine tuples with the same product and projective factors differ by a unique product-one scaling. For every a , the principal ideal (g_a) is T -stable, so its zero subscheme descends along the fpqc torsor q to a closed subscheme $Z_a \subset Z^{\text{sf}}$ (The Stacks Project Authors 2026, tags 04TW and 023T). The dense open U^{sf} of the affine source cannot be contained in the zero locus of the nonzero polynomial g_a ; faithful flatness therefore makes Z_a proper.

Moreover, Z^{sf} is irreducible: it is the nonzero tautological-line pullback over the irreducible square-free open in $\prod_i \mathbb{P}(V_{d_i})$. Hence $\dim Z_a < \dim Z^{\text{sf}}$. Since π is finite, $\pi(Z_a)$ is a proper closed subset of V_D^{sf} . Thus

$$V_D^{\text{good}} = V_D^{\text{sf}} \setminus \bigcup_{a=1}^t \pi(Z_a)$$

is dense and open, and every g_a is nonzero on every factorisation branch above it.

Write

$$B_0 = V_D^{\text{good}}, \quad Z_0 = \pi^{-1}(B_0), \quad P_0 = q^{-1}(Z_0), \quad G = (\mathbb{G}_m)^t.$$

Every g_a is invertible on P_0 , and equivariance gives a morphism $g : P_0 \rightarrow G$. Define

$$\theta = (q, g) : P_0 \longrightarrow Z_0 \times G.$$

Then the restricted normalised factorisation map factors as

$$\Phi_0 = (m_d, g) : P_0 \xrightarrow{\theta} Z_0 \times G \xrightarrow{\pi \times \text{id}_G} B_0 \times G. \quad (8.2)$$

Base-change θ by the fpqc cover $q \times \text{id}_G : P_0 \times G \rightarrow Z_0 \times G$. The torsor identity $P_0 \times_{Z_0} P_0 \cong P_0 \times T$ identifies the pullback with the Cartesian square

$$\begin{array}{ccc} P_0 \times T & \xrightarrow{\beta} & P_0 \times G \\ \downarrow a & & \downarrow q \times \text{id}_G \\ P_0 & \xrightarrow{\theta} & Z_0 \times G, \end{array} \quad (8.3)$$

where

$$a(x, \lambda) = \lambda x, \quad \beta(x, \lambda) = (x, g(x)\chi_W(\lambda)).$$

The square is Cartesian because two points in one q -fibre differ by a unique $\lambda \in T$, and $g(\lambda x) = \chi_W(\lambda)g(x)$. Smith normal form and torus automorphisms reduce χ_W to

$$(z_1, \dots, z_t) \mapsto (z_1^{s_1}, \dots, z_t^{s_t}). \quad (8.4)$$

On coordinate rings each power map is free of rank s_i , with basis $1, z_i, \dots, z_i^{s_i-1}$. Hence β is finite locally free of rank $|\det W|$. Finite local freeness and constant rank descend fpqc-locally (The Stacks Project Authors 2026, tag 02VO), so θ has the same rank. The second map in (8.2) is finite étale of rank ν ; finite locally free morphisms are stable under composition (The Stacks Project Authors 2026, tag 02K9). Thus Φ_0 is finite locally free of rank $\nu|\det W|$, proving total scheme degree without counting geometric points.

For (8.4), the function-field extension in coordinate i has total degree s_i , separable degree s'_i , and inseparable degree p^{a_i} . The root-partition cover is étale. Composition therefore gives the displayed total and separable degrees and the inseparable factor $p^{\sum_i a_i}$. Scheme-theoretically, $\ker \chi_W \cong \prod_i \mu_{s_i}$ has length $\prod_i s_i$, while its geometric-point count is $\prod_i s'_i$. Finally, Corollary 5.2 gives on X°

$$\det D\Phi = \pm \Delta_d \det(W) \prod_a g_a.$$

All factors except $\det W$ are units there, so the Jacobian criterion gives the asserted étaleness condition; equivalently, étaleness is fpqc-local (The Stacks Project Authors 2026, tag 02VN). $\square$

8.2 Full-unit, edge, and polynomial-monomial charts

Choose a $\mathbb{Z}$ -basis $\ell_1, \dots, \ell_t$ of the complete regular-unit character lattice L . Expressing each ℓ_a as an integral combination of edge characters produces Laurent units

$$u_a = \prod_{i < j} R_{ij}^{z_{a,ij}} \in \Gamma(U, \mathcal{O}_U)^\times.$$

Their square character matrix has determinant $[X^*(T) : L] = g^{k-1}h(\mathbf{e})$. Theorem 8.1 therefore gives the Laurent-unit normalisation the generic degree

$$g^{k-1}h(\mathbf{e}) \frac{D!}{\prod_i d_i!}. \quad (8.5)$$

This realises the full-list index on the pairwise-coprime open, but negative exponents need not extend across its boundary. By contrast, a set H of t individual resultants has torus degree $|\det W_H| = m(H)$, while nonnegative products of resultants correspond to nonnegative integral combinations of

the rows. Arbitrary polynomial semi-invariants form a still larger category and may require deletion of additional divisors before becoming units.

The primitive vector $(1, 2, 2)$ makes the distinction concrete. In edge order 12, 13, 23,

$$W_E = \begin{pmatrix} 2 & 1 \\ 1 & -1 \\ -2 & 0 \end{pmatrix},$$

whose maximal minors have absolute values 3, 2, 2. Hence $h = 1$, but no pair of individual resultants is unimodular. Nevertheless the nonnegative monomials

$$u_1 = R_{12}R_{13}R_{23}, \quad u_2 = R_{12}R_{23}$$

have character matrix

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} W_E = I_2. \quad (8.6)$$

Here $D = 5$ and $\nu = 5!/(1!2!2!) = 30$. Thus:

Normalising functions	Torus degree	Total generic degree
(R_{12}, R_{13})	3	90
(R_{12}, R_{23})	2	60
(R_{13}, R_{23})	2	60
(u_1, u_2)	1	30

The last chart is étale on $D(\Delta)$ in every characteristic. It proves that full-list saturation need not be realised by an edge basis and that a polynomial monomial chart can outperform every edge basis. It does not prove that every saturated resultant-character semigroup has a nonnegative unimodular basis.

For $k = 2$ and degrees $(1, 2)$, a rank-one normalising semi-invariant of unit character determinant gives generic degree 3, the number of ways to choose one of three roots for the labelled linear factor. For three equal linear factors, a tree of resultant normalising functions can have $|\det W| = 1$, while the root-partition factor is $3! = 6$. Neither observation produces a polynomial Keller map: global affine-slice recognition is an additional requirement.

9. Exact computational evidence

The supplied programs test the formulas over exact integer polynomial rings. They do not use floating point or random sampling in their default tiers. Their purpose is to detect convention, sign, exponent, and implementation errors in the stated identities.

The first multifactor implementation uses SymPy 1.14 and Berkowitz determinants. It constructs the coefficient differential directly from shifted products of complementary factors, constructs each pairwise Sylvester resultant independently, and compares every Plücker coordinate with the predicted torus minor. Its seven default degree vectors are

$$(1, 1), (1, 2), (1, 1, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (1, 1, 1, 1).$$

These cases comprise 143 symbolic Plücker coordinates. Five deliberate mutations omit one resultant, square the collision product, reverse a kernel row, reverse the recursive sign, or perturb the multiplication differential. All are detected.

The second implementation uses python-flint 0.9, multivariate integer polynomials, and fraction-free Bareiss determinants. It does not import the SymPy implementation. Its default tier checks 134 symbolic Plücker coordinates for three- and four-factor cases. It also computes the full square bordered Jacobian for the normalising functions (R_{12}, R_{23}) at degrees $(1, 1, 1)$ and $(1, 1, 2)$, whose character determinants are respectively one and two. Four structural mutations are detected.

A third program operates on exact integer character matrices. Through five vertices and degrees at most three, it checks 31,761 spanning-tree formulas, 52,741 general graph-minor formulas, and 351 Smith-form/bad-prime cases. Four negative controls substitute total degree for bipartite balance, omit the odd-cycle factor two, omit the common gcd in the Smith form, or extrapolate the two-factor equal-degree collapse to three factors.

A fourth Stage-4 program targets the strengthened Smith theorem. It constructs the complete character matrix independently, computes its Smith form over $\mathbb{Z}$, compares the last invariant with the symmetric formula, and checks the local annihilator valuation for every admissible pivot and every prime in its bounded domains. A separate tier enumerates actual spanning trees by Prüfer sequences and takes the gcd of their determinants. The default run contains 42,727 local/global Smith comparisons, 1,485 tree-gcd comparisons, and eight displayed-example checks, for 44,220 positive comparisons. Five negative controls detect omission of the factor two, replacement of valuations by prime support, conflation of full-list saturation with an edge basis, and omission of nonprimitive content.

Tier	Positive comparisons	Negative controls	Script SHA-256 prefix	Receipt SHA-256 prefix
SymPy	143 coordinates	5	3806a7c9b4bf	27d4ea5a3e89
multifactor				
FLINT	134 coordinates + 2	4	e7de053c910d	f352edfd686e
multifactor/ borders	borders			
Character lattice	84,853 comparisons	4	b91d2c7b0f11	5efd7944b92c
Local Smith theorem	44,220 comparisons	5	09ce5ad767d8	4d437bf2a6c8

The package manifest and Stage-1 provenance record contain the complete 64-character digests; the table uses unambiguous prefixes for legibility.

Across the four tiers there are 129,352 positive comparisons and 18 detected negative controls. Each default tier was run under ordinary Python and `python -O`. The normal and optimized receipts are byte-identical. This guard matters because proof checks implemented only with Python `assert` statements can disappear under optimization.

The two polynomial programs use distinct expression engines and determinant algorithms, which reduces the probability of a backend-specific error. They still implement the same producer specification and were written within the same project. Their agreement is cross-implementation evidence, not independent reproduction. Exact computation also does not prove novelty or replace the integral argument.

10. Antecedents, limitations, and the remaining programme

10.1 Contribution boundary

The square multiplication-Jacobian identity is a direct antecedent, not a novelty claim. Artin identifies the two-factor product-equation Jacobian with the transposed resultant matrix (Artin 2022, sec. 1.8, Lemma 1.8.5); Basu, Pollack, and Roy state the monic version (Basu et al. 2006, sec. 4.2.1); and Bhargava, Cremona, Fisher, and Gajović give monic-times-arbitrary multiplication over an arbitrary ring (Bhargava et al. 2022, Lemma 2.6). Chipalkatti supplies the corresponding projective tangent injectivity criterion (Chipalkatti 2003, Lemma 5.7). Chaperon and López de Medrano analyse the singularities of polynomial multiplication and prove the arbitrary-factor monic corank and pairwise-coprimality criterion (Chaperon and López de Medrano 2009, Theorems 1 and 3).

The product of pairwise resultants as a multifactor scalar is likewise a direct antecedent: Mahatab and Sampath’s polynomial CRT theorem gives it for monic factors (Mahatab and Sampath 2015, Theorem A.3). Chardin’s Koszul-complex treatment places the resultant in generalized Sylvester determinantal ideals (Chardin 1993, sec. IV, Lemma 1 and Remark 1). The determinant-of-complexes principle is classical (Jouanolou 1991; Demazure 2012; Gelfand et al. 1994).

The projective multiplication map, finite factorisation fibres, and labelled root partitions are likewise established background (Breiding et al. 2024). Kurth’s use of ordered linear factors modulo a scaling torus and the symmetric group is a close quotient-space antecedent (Kurth 1997). The signed support, rank, and powers of two in Theorem 6.1 are classical all-negative signed-incidence facts (Zaslavsky 1982, 1983; Grossman et al. 1995). The equal-weight vectors $\varepsilon_i + \varepsilon_j$ also occur in the arithmetic of classical signed root systems (Ardila et al. 2015, sec. 4.2), although the ambient lattice and the arbitrary cross-weighting here differ. Hanusa and Zaslavsky’s complete-incidence result concerns the least common multiple of all minors of a Kronecker-product matrix, not the gcd of maximal minors or Smith module of the present list (Hanusa and Zaslavsky 2011). Graph critical groups provide useful Smith-form context (Lorenzini 2008), but the present cokernel is not a reduced Laplacian cokernel and no chip-firing or monodromy-pairing interpretation is asserted.

Arithmetic multiplicity is likewise established language. For a represented list, $m(E)$ is its saturation index and obeys the GCD rule; over the complex torus it counts connected components of the common character kernel (Moci 2012; D’Adderio and Moci 2013). Matroids over $\mathbb{Z}$ and over DVRs retain the quotient module and local invariant sequence rather than only its order (Fink and Moci 2016). Accordingly, neither $h = m(E)$, the determinant gcd, nor the toric component interpretation is claimed as new.

The expanded audit therefore changes the emphasis. Theorem 4.1 is an explicit universal non-monic affine lift of classical square and monic identities, and Corollary 5.1 is its formal Cauchy – Binet contraction. Theorem 6.1 is a factorisation-specific cross-weighted application of classical signed-incidence theory. The strongest candidate result is the package formed by Theorems 7.1, 7.3, and 7.4 and Corollaries 7.2 and 7.5: the one-generator local module, every exact valuation, the symmetric global formula, spanning-tree-gcd equality, and the scaled Smith form for the complete cross-weighted list in the diagonal quotient. The regular-unit interpretation is specific to the factorisation open. Theorem 8.1 is best regarded as a scheme-theoretic synthesis of classical labelled root partitions with the standard degree of a torus isogeny.

Within the expanded bounded corpus, we did not locate the exact all-factor affine Plücker tensor with its orientation, the local presentation or closed index formula for this cross-weighted diagonal-quotient list, or the tree-gcd theorem in this setting. This is a bounded negative search result, not proof of priority. Equivalent results may exist in determinant-complex, signed-graphic, arithmetic-matroid, toric-arrangement, weighted-incidence, or representation language. Independent specialist search and proof reproduction remain necessary; no absolute priority claim is made.

10.2 Mathematical limitations

Equation (7.1) classifies regular units only on the universal pairwise-coprime open over $\mathbb{Z}$, and on its polynomial-ring analogue over a field. Under arbitrary base change, resultants may split and the base may acquire new units. Arbitrary polynomial semi-invariants are not classified, and they may become units after deleting additional coefficient divisors.

The full unit-character index is always realised by a square Laurent-unit basis. Saturation does not by itself give a basis made from individual resultants or nonnegative resultant monomials. The example $(1, 2, 2)$ has a nonnegative unimodular monomial basis, but the corresponding semigroup-basis problem is open in general. Likewise, the raw rectangular all-edge map has degree $[M : L]$ onto each branch-labelled image; possible identifications of different branch translates in an unlabelled rectangular target are not classified.

The numerical corank and pairwise-coprime regularity criterion for monic multifactor multiplication are already described by Chaperon and López de Medrano (Chaperon and López de Medrano 2009, Theorem 3). What remains untreated here is therefore not the set-theoretic rank classification, but the scheme-theoretic structure of higher collision strata: their Fitting ideals, multiplicities, embedded components, subresultant descriptions, and behaviour under arbitrary base change. The maximal-minor ideal above handles only the generic rank-loss divisor and its generic height-one multiplicity.

Most importantly, local determinant control does not solve the global affine-slice problem. A full-rank character matrix fixes the scaling torus up to a finite group scheme and can make a normalised factorisation chart étale off $\Delta = 0$. To obtain a polynomial Keller map on affine space one must also prove that a selected level set is isomorphic to affine space and that the restricted multiplication map has the required global fibre or properness behaviour. No such recognition theorem follows from the determinant line.

10.3 Assurance limitations

The parent two-factor theorem is an immutable unrefereed research candidate (Anonymous 2026). The parent venue and this extension share a producer/research process. The present induction has not been independently reproduced or formally verified. The source search is bounded, the programs are producer-written, and the manuscript has not undergone external peer review. These limits apply even if every local build and exact check passes.

10.4 High-value next problems

The immediate next mathematical tasks are narrower than the programme's largest aspirations.

1. Obtain an independently written proof of the affine Plücker lift, including every orientation convention, and independently reproduce the local Smith presentation and global formula.
2. Formalise the determinant-line composition lemma, the local Tietze elimination, and the Fitting-ideal consequences in a proof assistant.
3. Classify when the unit-character lattice has a basis of nonnegative resultant monomials, then investigate arbitrary semi-invariant vertical Jacobians after additional divisor deletions.
4. Refine the classical numerical corank formula to scheme-theoretic higher collision strata using subresultants and Fitting ideals.
5. Determine whether the complete unit-character group controls monodromy or compactification data beyond the generic torus chart.
6. Treat affine-slice recognition as a separate global geometry problem before drawing Keller or Hessian conclusions.

The most credible programme-level target is a classification of factorisation-derived étale charts and the isolation, if true, of exceptional affine-space slices. A four-dimensional Hessian application remains a conditional long-range problem, not a consequence suggested by the present proof alone.

11. Conclusion

The strongest candidate calculation in this paper is now exact. Theorem 7.1 reduces the complete primitive resultant-character quotient at every prime to one generator with the relations D_a and $2e_i e_j$. Corollary 7.2 gives every valuation of its order; Theorem 7.3 assembles those valuations into a symmetric global formula; and Theorem 7.4 shows that spanning-tree minors already determine the full arithmetic-matroid multiplicity. Scaling then gives the complete nonprimitive Smith form and the connected/étale decomposition of the unit-character residual group.

The route to this calculation begins with a classical multiplication Jacobian. Theorem 4.1 packages that antecedent as the exact affine determinant-line identity

$$\text{maximal-minor tensor} = \text{pairwise resultant divisor} \times \text{torus Plücker tensor},$$

with an explicit integral orientation. Multi-bordering pairs the torus tensor with the vertical Jacobian of the chosen normalising functions. For semi-invariants, this pairing is an integer character determinant. Pairwise resultants turn the character calculation into cross-weighted signed-incidence theory, from which the graph-minor formula follows. Standard arithmetic-matroid language then identifies the complete-list gcd as $m(E)$, while the local presentation evaluates that standard invariant and its full quotient module for this factorisation-derived list.

The rank-one degree difference is therefore the rank-one character determinant inside a broader Smith theory. In higher torus rank, a selected square character determinant controls a particular finite chart, whereas the complete unit-character lattice controls the residual group intrinsic to the fixed pairwise-coprime open. Theorem 8.1 translates square Smith data into total, separable, and inseparable scheme degrees through a finite-locally-free factorisation. These distinctions are the durable conclusion. The publication boundary remains provisional until the proofs and strengthened literature position receive genuinely independent specialist review.

Data and code availability

The research package contains the manuscript source, bibliography, exact SymPy and FLINT verification programs, the integer character-lattice program, normal and optimized receipts, the Stage-4 local-Smith verifier and receipts, a manifest, the proof and literature audit notes, and a formalisation blueprint. The package is a local research-stage child artefact at the time of writing. No public repository, archival deposit, DOI, or independent reproduction for this extension is asserted.

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Ethics declaration

This work is theoretical mathematics and involved no human participants, personal data, animals, clinical intervention, or sensitive field data. The principal research-ethics risks are attribution error, novelty overstatement, and inflation of producer checks into independent verification; the manuscript states explicit controls for each.

Author contributions

Conceptualization: [to be supplied]. **Formal analysis:** [to be supplied]. **Methodology:** [to be supplied]. **Software:** [to be supplied]. **Validation:** [to be supplied]. **Investigation:** [to be supplied]. **Writing – original draft:** [to be supplied]. **Writing – review and editing:** [to be supplied].

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Appendix A. Orientation and indexing conventions

1. Coefficients are ordered by factor, then from X^{d_i} to Y^{d_i} .
2. Sylvester rows are shifts of the earlier factor followed by shifts of the later factor.
3. Deleted source columns use zero-based indices.
4. The Plücker sign is $(-1)^{\sum I}$.
5. The torus basis is $e_b - e_k$, ordered by $b = 1, \dots, k - 1$.
6. Gradient rows follow multiplication rows and are ordered $g_1, \dots, g_{k-1}$.
7. The multi-border Laplace sign is $(-1)^{nt + \binom{t}{2}}$.

Changing any convention changes intermediate signs. The invariant content is the determinant-line equality after orientations are transformed together.

Appendix B. Verification boundary

The default receipts verify the exact cases listed in Section 9 and include deliberate negative controls. Larger exploratory cases were also tested but are not part of the frozen default receipt. No finite list of symbolic cases proves the universal theorem. The proof is the induction in Sections 3 and 4; the programs are regression tests for its coordinate consequences.

Appendix C. Formalisation outline

A proof-assistant kernel can be divided into the following layers:

1. coefficient convolution and the universal multiplication matrix;
2. the self-contained two-factor complementary-minor identity of Lemma E.1;
3. signed complementary minors and the block-composition lemma;
4. resultant multiplicativity and the all-factor induction;
5. Laplace expansion, Cauchy – Binet, and multi-border contraction;
6. integer character matrices and Smith determinantal divisors; and
7. the weighted incidence reduction.

The universal polynomial identities should be formalised before geometric language about divisors or étaleness. This order keeps the trusted kernel small and isolates the sign-sensitive finite-dimensional linear algebra.

Appendix D. Bounded antecedent and claim matrix

The following matrix records the source-to-claim boundary used in this draft. It is a bounded map of the examined corpus, not a proof that no closer antecedent exists.

Object in this paper	Closest examined antecedent	What the antecedent establishes	Candidate increment evaluated here
Square two-factor product Jacobian	Artin, Basu – Pollack – Roy, and Bhargava – Cremona – Fisher – Gajović (Artin 2022; Basu et al. 2006; Bhargava et al. 2022)	The coefficient-product Jacobian is a Sylvester matrix, hence the resultant, including a monic-times-arbitrary identity over an arbitrary ring	Retain every affine coefficient direction and record the complete complementary-minor vector with a fixed integral orientation
Projective and monic multifactor rank	Chipalkatti and Chaperon – López de Medrano (Chipalkatti 2003; Chaperon and López de Medrano 2009)	Projective tangent injectivity for coprime factors, and the monic arbitrary-factor corank and pairwise-coprimality criterion	Refine the generic rank-loss set to an exact universal maximal-minor ideal with generic height-one multiplicities

Object in this paper	Closest examined antecedent	What the antecedent establishes	Candidate increment evaluated here
Scalar product $\prod_{i < j} R_{ij}$	Mahatab – Sampath (Mahatab and Sampath 2015)	Determinant of the polynomial Chinese-remainder map for monic factors	Restore all affine leading-coefficient directions and identify the full complementary-minor tensor
Resultant divisibility in determinantal systems	Chardin (Chardin 1993)	Resultant factors and generic gcd behaviour for generalized Sylvester/Koszul minors	Give every signed maximal minor explicitly for the coefficient differential of labelled multiplication
Determinant-of-complexes framework	Jouanolou, Demazure, and GKZ (Jouanolou 1991, 1997; Demazure 2012; Gelfand et al. 1994)	General resultant constructions, inertia forms, and discriminantal determinants	Isolate the torus-kernel Plücker line and its border contraction in this affine factorisation map
Two-factor affine identity	Parent candidate (Anonymous 2026)	Rank-one complementary-minor and bordered identities	Prove the all-factor induction, multi-border formula, and higher-rank character interpretation
Projective multiplication fibres	Breiding – Kohn – Sturmfels (Breiding et al. 2024)	Projective polynomial multiplication and finite root-partition geometry	Give an fpqc-local finite-locally-free factorisation that composes the root-partition cover with a translated torus isogeny
Linear-factor quotient geometry	Kurth (Kurth 1997)	Ordered linear factors modulo scaling and permutation actions	Retain arbitrary positive degrees and compute the resultant-character lattice integrally
Signed support of character bases	Zaslavsky with erratum, Grossman – Kulkarni – Schochetman, and Hessert – Mallik (Zaslavsky 1982, 1983; Grossman et al. 1995; Hessert and Mallik 2023)	All-negative signed-incidence rank, dependence, odd-cycle support, powers of two, and odd-unicyclic invertibility	Evaluate the appended degree balance and cross-degree product and identify the characters with pairwise resultants

Object in this paper	Closest examined antecedent	What the antecedent establishes	Candidate increment evaluated here
Multiplicity of the complete character list	Moci and D’Adderio – Moci (Moci 2012; D’Adderio and Moci 2013)	Saturation multiplicity, GCD rule, and complex toric component count	Give a closed evaluation for the complete cross-weighted diagonal-quotient list and its factorisation-open interpretation
Exact full-set quotient module	Fink – Moci; Ardila – Castillo – Henley as an adjacent equal-weight root-list case (Fink and Moci 2016; Ardila et al. 2015)	Matroid-over-ring/ DVR framework and arithmetic multiplicities for classical signed-root lists	Prove the one-generator local presentation, every valuation, the scaled global Smith form, and connected/étale residual group factors
Complete-graph incidence subdeterminants	Hanusa – Zaslavsky (Hanusa and Zaslavsky 2011)	Least common multiple of all minors of a fixed Kronecker-product complete-incidence matrix	Compute a gcd of maximal minors and the full Smith module for a different cross-weighted signless list in the diagonal quotient
Jacobian ideals of binary discriminants	D’Andrea – Chipalkatti (D’Andrea and Chipalkatti 2007)	Structure and resolutions for coefficient-gradient ideals of binary discriminants and resultants	Study the rectangular differential of factor multiplication rather than the gradient ideal of the resultant hypersurface

The affine Plücker identity is best viewed as a derived, coordinate-complete refinement of the exact square antecedents. The strongest conditional originality claim supported by this matrix is narrower: the exact local module, closed valuation/global formula, and tree-gcd equality for the cross-weighted complete-edge list were not located in the bounded search. Specialist review could still identify an equivalent formulation in weighted-incidence, arithmetic-matroid, toric, logarithmic, or representation-theoretic language.

Appendix E. Self-contained affine lift of the classical two-factor Jacobian identity

Classical sources identify a square two-factor product Jacobian with the resultant (Artin 2022, sec. 1.8, Lemma 1.8.5; Basu et al. 2006, sec. 4.2.1; Bhargava et al. 2022, Lemma 2.6). This appendix supplies the coordinate-complete affine complementary-minor lift and fixes its sign in the conventions of this paper. It also reproduces the parent rank-one base theorem (Anonymous 2026), so Theorem 4.1 remains self-contained without asserting priority for the underlying square Jacobian identity.

Lemma E.1 (two-factor complementary minors over $\mathbb{Z}$). Let

$$A = \sum_{i=0}^r a_i X^{r-i} Y^i, \quad B = \sum_{j=0}^s b_j X^{s-j} Y^j, \quad r, s \geq 1,$$

and write $AB = \sum_{h=0}^{r+s} c_h X^{r+s-h} Y^h$. Order source coordinates as $a_0, \dots, a_r, b_0, \dots, b_s$ and target coordinates as $c_0, \dots, c_{r+s}$. Let M be the coefficient differential,

$$M[h, i] = b_{h-i}, \quad M[h, r+1+j] = a_{h-j},$$

with out-of-range subscripts equal to zero, and put

$$\kappa = (a_0, \dots, a_r, -b_0, \dots, -b_s).$$

Then

$$\boxed{(-1)^\ell \det M_\ell = (-1)^{r(s+1)} \operatorname{Res}(A, B) \kappa_\ell, \quad 0 \leq \ell \leq r+s+1.} \quad (\text{E.1})$$

Proof. Both sides are polynomials with integer coefficients, so it suffices to prove their equality on a nonempty Zariski-open subset over $\mathbb{C}$. Dehomogenise by $t = Y/X$, work where $a_r b_s \neq 0$, and write

$$a(t) = a_r \prod_{i=1}^r (t - \alpha_i), \quad b(t) = b_s \prod_{j=1}^s (t - \beta_j),$$

with all α_i, β_j pairwise distinct. In the Sylvester convention of Section 2,

$$\operatorname{Vand}(x_1, \dots, x_m) := \prod_{1 \leq u < v \leq m} (x_v - x_u).$$

Every Vandermonde below uses the displayed order of its arguments. With this orientation,

$$\operatorname{Res}(A, B) = a_r^s b_s^r \prod_{i,j} (\beta_j - \alpha_i) = b_s^r \prod_j a(\beta_j) = (-1)^{rs} a_r^s \prod_i b(\alpha_i). \quad (\text{E.2})$$

For completeness, transpose the Sylvester matrix and view it as $(u, w) \mapsto ua + wb$. Evaluating the output at $(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s)$ makes the matrix block anti-diagonal. Its determinant is

$$(-1)^{rs} \left(\prod_i b(\alpha_i) \right) \operatorname{Vand}(\alpha) \left(\prod_j a(\beta_j) \right) \operatorname{Vand}(\beta).$$

Division by the evaluation determinant

$$\operatorname{Vand}(\alpha) \operatorname{Vand}(\beta) \prod_{i,j} (\beta_j - \alpha_i)$$

gives (E.2); the specialisation $(A, B) = (X^r, Y^s)$ fixes the normalisation as $+1$.

Choose $\tau \in \mathbb{C}$ distinct from every root and evaluate the degree- $(r+s)$ target at

$$\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s, \tau.$$

Let V be this evaluation matrix. Its determinant is the Vandermonde in the displayed order. At α_i , the evaluated Jacobian row is

$$b(\alpha_i)(1, \alpha_i, \dots, \alpha_i^r \mid 0);$$

at β_j , it is

$$a(\beta_j)(0 \mid 1, \beta_j, \dots, \beta_j^s).$$

The τ -row is the sum of its A -block and B -block parts. After one source column is deleted, one summand has deficient block rank and vanishes.

We also need the gap-Vandermonde identity. For m variables $x_1, \dots, x_m$, let $W_{\hat{h}}(x)$ be the m -square evaluation matrix with exponents $\{0, \dots, m\} \setminus \{h\}$. Comparing coefficients of T^h in the Vandermonde determinant on $(x_1, \dots, x_m, T)$ gives

$$\det W_{\hat{h}}(x) = e_{m-h}(x) \text{Vand}(x). \quad (\text{E.3})$$

Suppose first that the deleted column is b_j , of global index $\ell = r + 1 + j$. The surviving term places the τ -row in the A -block. Moving it past the s beta rows contributes $(-1)^s$, and block expansion gives

$$\begin{aligned} \det(VM_{\hat{\ell}}) &= (-1)^s \left(\prod_i b(\alpha_i) \right) b(\tau) \text{Vand}(\alpha, \tau) \\ &\quad \cdot \left(\prod_j a(\beta_j) \right) \det W_{\hat{j}}(\beta). \end{aligned} \quad (\text{E.4})$$

Use (E.2) – (E.3),

$$b(\tau) = b_s \prod_j (\tau - \beta_j), \quad \prod_i b(\alpha_i) = (-1)^{rs} b_s^r \prod_{i,j} (\beta_j - \alpha_i),$$

and divide by $\det V$. All Vandermonde and τ -factors cancel, leaving

$$\det M_{\hat{\ell}} = (-1)^{s(r+1)} b_s e_{s-j}(\beta) \text{Res}(A, B). \quad (\text{E.5})$$

Since $b_j = (-1)^{s-j} b_s e_{s-j}(\beta)$,

$$\begin{aligned} (-1)^\ell \det M_{\hat{\ell}} &= (-1)^{r+1+j+s(r+1)+s-j} b_j \text{Res}(A, B) \\ &= (-1)^{r(s+1)} (-b_j) \text{Res}(A, B), \end{aligned}$$

which is (E.1) in the B -block.

If the deleted column is a_i , its global index is $\ell = i$. The surviving term places the τ -row in the B -block, and the existing row order is already block compatible:

$$\begin{aligned} \det(VM_{\hat{i}}) &= \left(\prod_i b(\alpha_i) \right) \det W_{\hat{i}}(\alpha) \\ &\quad \cdot \left(\prod_j a(\beta_j) \right) a(\tau) \text{Vand}(\beta, \tau). \end{aligned} \quad (\text{E.6})$$

The same cancellations yield

$$\det M_{\hat{i}} = (-1)^{rs} a_r e_{r-i}(\alpha) \text{Res}(A, B) = (-1)^{rs+r-i} a_i \text{Res}(A, B). \quad (\text{E.7})$$

Multiplication by $(-1)^i$ gives (E.1) in the A -block. The identity holds on the chosen dense open subset, hence as a polynomial identity over $\mathbb{Z}$. $\square$

For reference, the two deleted-column parity calculations are:

Deleted block	Global index	Sign of $\det M_{\hat{\ell}}$ after coefficient substitution		Cofactor sign	Kernel- coordinate sign	Remaining scalar sign
A , column a_i	i	$rs + r - i$		i	0 for a_i	$r(s + 1)$
B , column b_j	$r + 1 + j$	$s(r + 1) + s - j$		$r + 1 + j$	1 for $-b_j$	$r(s + 1)$

All entries in the last four columns are exponents of (-1) , read modulo two. This table records the cancellations between (E.4) – (E.5) and (E.6) – (E.7) without changing the proof.

Expansion along an arbitrary final row v now gives the equivalent bordered form

$$\det \begin{pmatrix} M \\ v \end{pmatrix} = (-1)^{s(r+1)+1} \text{Res}(A, B) \langle v, \kappa \rangle,$$

which also confirms the $k = 2$ specialisation of Corollary 5.1.

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